

TADI: Target-Aware Direct Injection with Dual-Path Target Modeling for Implicit Hate Speech Detection

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ABSTRACT

Implicit hate speech, which conveys hostility toward specific groups without using overtly offensive language, poses significant challenges for automatic detection. Identifying implicit hate typically requires modeling both the target of the attack and the contextual attitude toward that target. However, most existing detection methods rely on sentence-level representations and fail to explicitly capture target-context relationships. While recent target-aware approaches have demonstrated the value of incorporating target information, they largely depend on explicit target candidates or external identification tools, limiting their effectiveness when targets are expressed implicitly through pronouns, metaphors, or cultural allusions. To address these limitations, we propose TADI (Target-Aware Direct Injection), a framework that models target information through dual pathways: an implicit pathway that automatically discovers target-relevant representations via attention mechanisms, and an explicit pathway that leverages available target candidate positions. TADI computes target-context relationship vectors from both pathways and directly injects them into the classification representation through gated residual connections. Additionally, we introduce auxiliary training objectives, including target supervision and hard negative contrastive learning, to enhance the stability of target extraction and the discriminability of boundary samples. Experiments on four hate speech and offensive language datasets with two pre-trained encoders demonstrate that TADI achieves strong and generally superior performance compared to existing methods in both in-dataset and cross-dataset evaluation settings.

1. Introduction

With the rapid growth of social media platforms, online hate speech has become a critical issue affecting online ecosystems and platform governance [13, 24]. Hate speech can generally be categorized into two forms: explicit hate and implicit hate. Explicit hate typically contains abusive vocabulary or directly aggressive expressions, making it relatively easier to detect through keyword matching or text classification methods [26, 8]. In contrast, implicit hate speech often lacks overtly offensive words and instead conveys hostility toward specific groups through metaphors, sarcasm, stereotypes, or cultural codes [11, 31]. Due to the absence of explicit lexical cues, the automatic detection of implicit hate speech remains a significant challenge and constitutes an important open problem in current hate speech detection research.

From the perspective of discriminating hateful semantics, identifying implicit hate often requires simultaneously determining the target of the attack and the contextual attitude toward that target [4, 17]. For example, the sentence “These outsiders will never integrate into our community” contains no overtly abusive words, yet expresses hostility through group reference and exclusionary context. This indicates that implicit hate speech detection not only requires assessing whether the overall sentence semantics carry a negative orientation, but also necessitates modeling the target

of the text and the relationship between the context and the target. However, a substantial number of existing detection methods still primarily rely on sentence-level representation learning, typically using sentence-level representations as the main basis for classification [1, 15, 28], which makes it difficult to explicitly capture the aforementioned target-context relationships and tends to produce misclassifications when dealing with implicit hate speech. As illustrated in Fig. 1, such methods may fail to capture the relationship between an implicit target and its exclusionary context.

In recent years, researchers have begun to investigate the role of target information in hate speech detection. Some studies employ multi-task learning to jointly predict target categories [5], while others leverage adversarial training to mitigate spurious correlations introduced by target group features [27]. Among these, AmpleHate [17] obtains target information through explicit target identification, computes relationship vectors between the target and the context, and injects them into the classification representation, demonstrating the effectiveness of target-context relationship modeling for hate speech detection. However, such methods largely depend on explicit target candidates or external identification tools. When targets appear implicitly through pronouns, metaphors, or cultural allusions, these methods often struggle to obtain stable and comprehensive target information. Consequently, how to model implicit target information while reducing the reliance on external tools remains an open research question in hate speech detection.

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To address the aforementioned challenges, we propose TADI (Target-Aware Direct Injection), a framework designed for hate speech detection. TADI simultaneously models target information through two pathways—implicit and explicit—and directly injects the target-context relationships into the classification representation, enabling the model to explicitly leverage target-context relationships for classification (Fig. 1). Furthermore, we introduce auxiliary training mechanisms for target representations to improve the stability of target extraction and enhance the discriminability of boundary samples, i.e., semantically similar samples with different labels that are difficult to distinguish in the representation space. Experimental results on four hate speech and offensive language datasets with two pre-trained encoders demonstrate that TADI achieves generally superior detection performance compared to existing methods in the majority of settings. The main contributions of this paper are summarized as follows:

- We propose the TADI framework, which models target information through two complementary pathways, namely implicit target extraction and explicit target candidate aggregation. The framework injects target-context relationships into the classification representation to enhance the model's ability to discriminate implicit hate speech.
- We design auxiliary training mechanisms for target representations, incorporating target supervision and hard negative contrastive constraints to improve the stability of target extraction and target-aware representation learning.
- We conduct systematic experiments on three English and one Chinese hate speech and offensive language datasets, validating the effectiveness and applicability of TADI through both in-dataset evaluation and cross-dataset generalization evaluation.

2. Related Work

2.1. Hate Speech Detection

Early hate speech detection relied on lexical features and traditional machine learning methods. Waseem and Hovy [26] constructed classifiers based on character n-grams, and Davidson et al. [8] further delineated the boundary between hate speech and offensive language. With the advancement of pre-trained language models, encoder fine-tuning has become the dominant paradigm for this task, achieving strong performance on standard benchmarks. Meanwhile, implicit hate has emerged as a more challenging subproblem. ElSherief et al. [11] constructed the Implicit Hate Corpus and systematically defined a taxonomy of implicit hate categories, providing a foundation for studying expressions that lack overt offensive words. Zeng et al. [31] investigated metaphorical forms of implicit hate expression and showed that large language models frequently misclassify such figurative hate as harmless. Furthermore, Yuan and

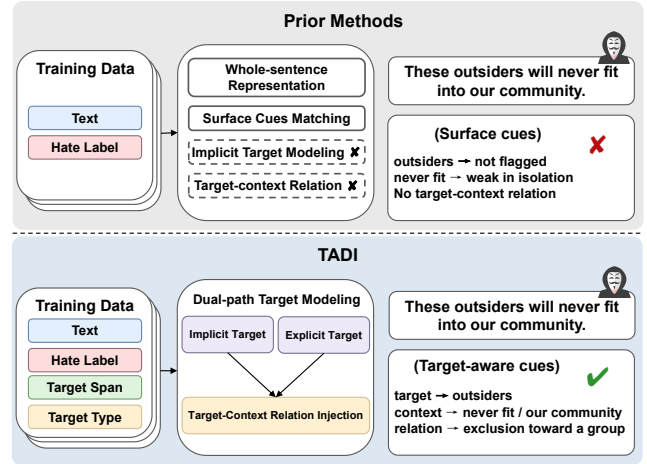


Figure 1: Comparison of surface-cue based detection and TADI for implicit hate speech detection. The upper panel shows detection based on whole-sentence representation and surface cues, while the lower panel shows TADI based on dual-path target modeling and target-context relation injection.

Rizoiu [28] demonstrated through cross-dataset evaluation that models trained on a single dataset often fail to generalize to other data sources. Overall, existing methods largely rely on sentence-level representations, lacking explicit modeling of target information and its contextual relationship.

2.2. Target-Aware Hate Speech Detection

Several recent studies have incorporated target information into hate speech detection. Chiril et al. [5] adopted a multi-task learning framework that jointly predicts target categories during detection, learning the association between textual toxicity and the target of attack. Mathew et al. [21] proposed HateXplain, which provides target community and rationale span annotations for each instance, establishing a data foundation for target-oriented explainable detection. In the Chinese scenario, Bai et al. [2] introduced span-level target-aware toxicity extraction for fine-grained detection. Regarding target-related bias, Xia et al. [27] leveraged adversarial training to reduce spurious dependence on group features, and Chen et al. [4] investigated model generalization to unseen target groups from a target-aware fairness perspective. The most closely related work is AmpleHate [17], which identifies targets explicitly and injects target-context relationship vectors into the classification representation. However, AmpleHate depends on explicit target candidates, and its coverage is limited when targets appear implicitly through pronouns or metaphors. How to model implicit target information without relying on external tools remains an open problem.

2.3. Contrastive Learning and Hate Speech Detection

Supervised contrastive learning [14] enhances representational discriminability by pulling together same-class samples and pushing apart different-class samples in the representation space. Gunel et al. [12] extended this paradigm

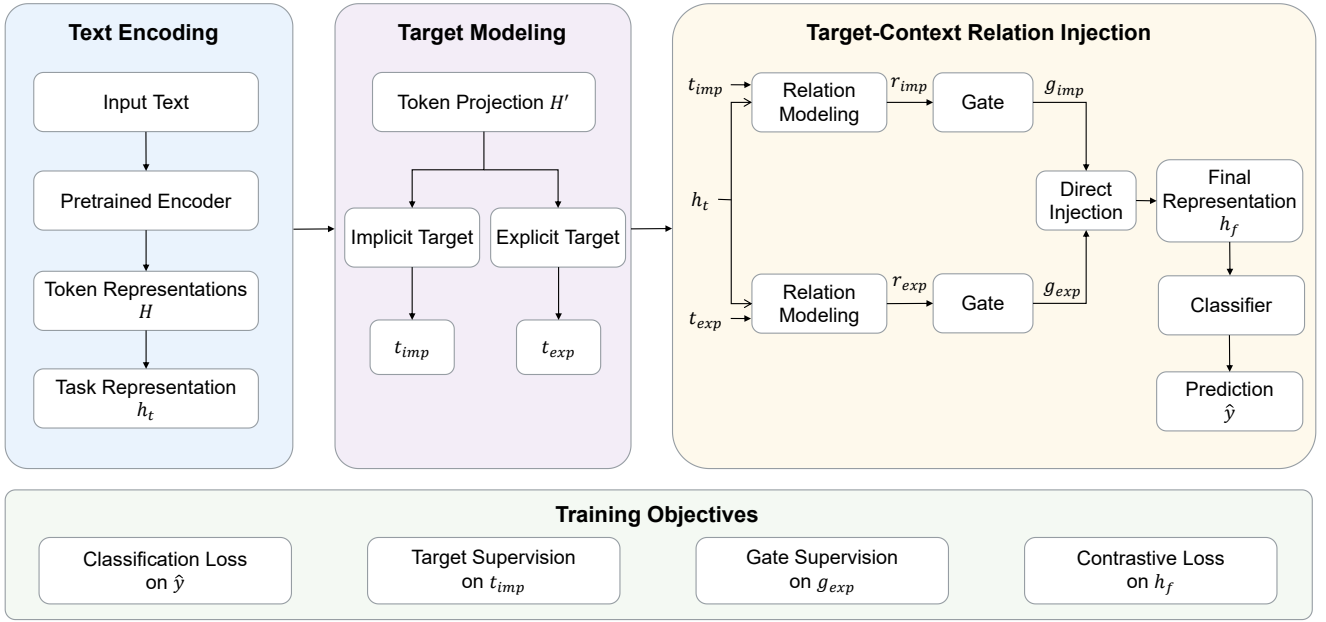


Figure 2: Overall architecture of TADI. The framework encodes the input text, models target information through implicit and explicit pathways, and injects target-context relation vectors into the task representation. The bottom panel summarizes the training objectives and their corresponding optimization targets.

to pre-trained language model fine-tuning, demonstrating improved robustness on downstream tasks. In hate speech detection, implicit hateful samples and non-hateful samples are often highly proximate in the semantic space, making it difficult for cross-entropy loss alone to distinguish them. Kim et al. [16] first applied contrastive learning to improve cross-dataset generalization of implicit hate classifiers. SharedCon [1] exploits shared semantics between explicit and implicit hate to construct contrastive objectives, enhancing the recognition of implicit expressions. LAHN [15] proposes a label-aware hard negative sampling strategy combined with momentum contrastive learning, targeting semantically similar but differently labeled sample pairs. These methods effectively enhance sentence-level representational discriminability; however, their positive and negative sample relationships are determined by label information alone, without exploiting the semantic information provided by target-aware representations.

3. Method

3.1. Overall Framework

We formulate hate speech detection as a binary classification task: given an input text x , the model predicts the label $y \in \{\text{hate}, \text{non-hate}\}$. The overall architecture of TADI is illustrated in Fig. 2. Its forward process primarily consists of target modeling and target-context relationship injection, with auxiliary training objectives introduced during the training phase.

Specifically, the model first obtains token-level representations and a sentence-level task representation through a pre-trained encoder. Subsequently, TADI models target information through two pathways: the implicit pathway

automatically discovers target representations relevant to the hate speech judgment from within the text via an attention mechanism, while the explicit pathway aggregates explicit target representations based on target candidate positions. On this basis, the model computes relationship vectors between the target and the context under each pathway, and injects them into the task representation through gated residual connections to obtain the final classification representation that incorporates target-context relationships. During the training phase, the model introduces target supervision, gate supervision, and hard negative contrastive constraints in addition to the primary classification loss, to improve the stability of target modeling and the discriminability of representations.

The dual-pathway design reflects the complementary nature of implicit and explicit target information. The implicit pathway does not require any external annotation or tool and is therefore applicable to all input samples, but it relies entirely on learned attention and may be less precise when target cues are subtle. The explicit pathway, when available, provides direct positional constraints that anchor the target representation to specific text spans, but its coverage is inherently limited by the completeness of candidate information. By combining both pathways, TADI can leverage explicit positional priors when they exist while maintaining robust target modeling through the implicit pathway when they do not. The gated residual injection mechanism further supports this design: the task representation h_t is preserved as the base, and target-context relationship information is added as a controlled increment. When target information is unreliable for a given sample, the gate values can suppress the injection magnitude toward zero, allowing the model

to gracefully degrade to the behavior of the base encoder without architectural modification.

3.2. Target Modeling

Text encoding. Given an input text x , the model first obtains token-level hidden states $H = \{h_1, h_2, \dots, h_L\} \in \mathbb{R}^{L \times d}$ through a pre-trained Transformer encoder [25], where L denotes the sequence length and d denotes the hidden dimension. On this basis, the hidden state h_c at the [CLS] position is passed through a task transformation layer to obtain the task representation:

$$h_t = \text{LayerNorm}(\text{GELU}(W_t h_c + b_t)), \quad (1)$$

where $W_t \in \mathbb{R}^{d \times d}$ and $b_t \in \mathbb{R}^d$ are learnable parameters. To effectively model target information while reducing the reliance on external tools, TADI obtains target representations from both implicit and explicit pathways. The two pathways share a token projection layer that maps the encoder outputs to token representations used for subsequent target modeling:

$$h'_i = W_p h_i, \quad H' = \{h'_1, h'_2, \dots, h'_L\}, \quad (2)$$

where $W_p \in \mathbb{R}^{d \times d}$ is a learnable projection matrix, h'_i denotes the projected representation of the i -th token, and H' is the projected token representation sequence.

Implicit target extraction. The goal of the implicit pathway is to automatically discover target information relevant to the hate speech judgment from within the text. Specifically, TADI generates a query vector from the task representation h_t and key vectors from the token representations:

$$q = W_q h_t, \quad k_i = W_k h'_i, \quad (3)$$

where $W_q, W_k \in \mathbb{R}^{d \times d}$ are learnable parameters. The model then computes the attention weights between the task representation and each token:

$$\alpha_i = \frac{\exp(q^\top k_i / (\sqrt{d} \tau))}{\sum_{j=1}^L \exp(q^\top k_j / (\sqrt{d} \tau))}, \quad (4)$$

where τ is a temperature coefficient that controls the sharpness of the attention distribution. Based on the attention weights, the model performs a weighted aggregation over the projected token representations to obtain the implicit target representation:

$$t_{\text{imp}} = \sum_{i=1}^L \alpha_i h'_i. \quad (5)$$

In this process, the query vector is derived from the task representation and the key vectors are derived from the token representations. Consequently, the attention weight α_i reflects the degree of relevance between the i -th token and the current hate speech judgment, enabling soft localization of potential target information within the text. This pathway

does not rely on external identification tools and is applicable to all input samples.

A key design choice is to derive the query from the sample-specific task representation h_t rather than from a globally shared learnable parameter. Since h_t encodes the overall semantics of the current input, using it as the query enables the attention mechanism to adaptively locate target-relevant tokens conditioned on the specific context of each sample. This is particularly important for implicit hate speech, where the same surface form (e.g., a group reference) may serve as a target in one context but not in another. A context-independent query would lack this adaptive capacity and may attend to group-related tokens indiscriminately, regardless of whether the surrounding context expresses hostility.

Explicit target candidates. The explicit pathway leverages target candidate positions to provide more direct positional information. Let $m \in \{0, 1\}^L$ be a token-level candidate mask, where $m_i = 1$ indicates that the i -th token belongs to a target candidate region. In practice, the candidate mask is constructed from available target-related annotations or category terms by matching them against the input text at the token level. The implicit pathway is applied to all samples regardless of candidate availability and therefore provides the primary source of target information, while the explicit pathway offers complementary positional constraints when candidates can be derived. The explicit target representation is obtained by average pooling over the token representations at candidate positions:

$$t_{\text{exp}} = \frac{\sum_{i=1}^L m_i h'_i}{\sum_{i=1}^L m_i}. \quad (6)$$

When no target candidate is available for a given sample, the candidate mask satisfies $\sum_{i=1}^L m_i = 0$. In this case, the explicit target representation t_{exp} is set to a zero vector, and the corresponding gate value g_{exp} is driven toward zero through gate supervision (described in Section 3.4), effectively deactivating the explicit pathway for that sample. This design ensures that the model does not require a separate conditional branch to handle the absence of candidates; instead, the gating mechanism provides a unified and differentiable way to modulate the contribution of the explicit pathway based on candidate availability.

Compared with the implicit pathway, the explicit pathway does not need to re-learn the attention distribution over targets but instead directly aggregates target representations using candidate positions. Therefore, it can provide a more definitive positional prior for the model, serving as a complement to the implicit pathway.

Relationship between the two pathways. The implicit pathway serves as the primary pathway for TADI to obtain target information, capable of automatically modeling potential target information from within the text. The explicit pathway provides more direct positional constraints when candidate information is available. The representations t_{imp} and t_{exp} obtained from the two pathways will respectively

participate in the subsequent target-context relationship computation and jointly contribute to enhancing the final classification representation.

3.3. Target-Context Relationship Injection

After obtaining the target representations, TADI further models the semantic relationship between the target and the textual context, and injects this relationship information into the task representation to assist classification. In this paper, the task representation h_t is used to represent the contextual semantics of the text. The implicit and explicit pathways adopt the same relationship computation and injection structure, and operate independently. The following description uses the implicit pathway as an illustrative example.

Relationship vector computation. To capture the multi-angle interaction between the task representation h_t and the implicit target representation t_{imp} , the model concatenates four combinations of the two vectors and feeds them into a relation network:

$$z_{\text{imp}} = [h_t; t_{\text{imp}}; h_t \odot t_{\text{imp}}; h_t - t_{\text{imp}}], \quad (7)$$

$$r_{\text{imp}} = \text{MLP}_r(z_{\text{imp}}), \quad (8)$$

where $[\cdot]$ denotes vector concatenation, $\odot$ denotes element-wise multiplication, and $z_{\text{imp}} \in \mathbb{R}^{4d}$. This four-way combination follows the matching heuristic widely adopted in natural language inference [3], which has proven effective for capturing multi-faceted semantic relationships between two representations. Specifically, the element-wise product $h_t \odot t_{\text{imp}}$ captures the degree of semantic alignment between the target and its context, where high values indicate that the two representations share similar features in certain dimensions. The element-wise difference $h_t - t_{\text{imp}}$ captures the semantic gap between the target and context, highlighting dimensions where the two diverge. Together with the original representations h_t and t_{imp} , these four terms provide complementary perspectives—identity, correspondence, alignment, and divergence—enabling the relation network MLP_r to learn a comprehensive target-context relationship vector $r_{\text{imp}} \in \mathbb{R}^d$.

Gated residual injection. Different samples exhibit different degrees of dependence on target information. To this end, the model introduces a gating mechanism to adaptively regulate the injection strength of relationship information:

$$g_{\text{imp}} = \sigma(\text{MLP}_g(z_{\text{imp}})), \quad (9)$$

where MLP_g is a gating network independent of the relation network, σ is the sigmoid function, and $g_{\text{imp}} \in [0, 1]$ is a scalar gate value. A larger gate value indicates that the classification of the current sample requires greater participation of target-context relationship information. The explicit pathway adopts the same structure. Specifically, replacing t_{imp} with t_{exp} in the above equations yields the explicit pathway's relationship input z_{exp} , relationship vector r_{exp} , and gate value g_{exp} .

We adopt a scalar gate rather than a vector-valued gate for two reasons. First, a scalar gate applies a uniform scaling

factor to the entire relationship vector, which reduces the risk of dimension-level instability that may arise when individual dimensions are gated independently during early training. Second, scalar gating yields a single interpretable value per sample per pathway, making it straightforward to analyze which samples rely more heavily on target-context relationship information and to apply direct supervision through the gate supervision loss.

Direct injection. The relationship vectors from both pathways are weighted by their respective gates and added to the task representation in a residual form, yielding the final classification representation that incorporates target-context relationships:

$$h_f = h_t + \lambda_{\text{imp}} \cdot g_{\text{imp}} \cdot r_{\text{imp}} + \lambda_{\text{exp}} \cdot g_{\text{exp}} \cdot r_{\text{exp}}, \quad (10)$$

where λ_{imp} and λ_{exp} are preset scaling coefficients that control the injection magnitude of the two pathways. The residual addition enables the model to incorporate relationship information from both the implicit and explicit target pathways while preserving the overall semantic information in h_t . Finally, h_f is passed through a linear classification layer to produce the prediction:

$$\hat{y} = \text{softmax}(W_c h_f + b_c), \quad (11)$$

where W_c and b_c are the classification layer parameters, and $\hat{y}$ is the model's predicted probability distribution over hate/non-hate.

3.4. Training Objectives

The training objective of TADI consists of a primary classification loss and three types of auxiliary losses.

Primary classification loss. Based on the predicted distribution $\hat{y}$ and the ground-truth label y , we adopt the standard cross-entropy loss to optimize the binary classification of hate/non-hate:

$$\mathcal{L}_{\text{cls}} = \text{CE}(\hat{y}, y). \quad (12)$$

This loss directly constrains the model's final predictions and serves as the primary optimization objective during training.

Target supervision loss. To improve the reliability of implicit target modeling, we supervise the target representation from two aspects: attention alignment and category semantics. First, annotated target position information is used to construct a token-level target distribution, which constrains the attention weights α in Eq. (4) to concentrate on target-related tokens, guiding the model to locate target information within the text. Second, the implicit target representation t_{imp} is fed into a linear classification layer to predict the target category, guiding t_{imp} to encode the category semantics of the target. The above attention alignment and category prediction jointly constitute the target supervision loss:

$$\mathcal{L}_{\text{tar}} = \mathcal{L}_{\text{att}} + \mathcal{L}_{\text{type}}, \quad (13)$$

where $\mathcal{L}_{\text{att}}$ denotes the attention alignment loss and $\mathcal{L}_{\text{type}}$ denotes the target category prediction loss.

Gate supervision loss. To keep the gate strength of the explicit pathway consistent with the availability of target candidates, we apply supervision to the explicit gate value g_{exp} . Specifically, for samples where explicit target candidates are available ($\sum_{i=1}^L m_i > 0$), the supervision target is set to 1, encouraging the model to increase the participation of the explicit pathway; for samples without candidates ($\sum_{i=1}^L m_i = 0$), the target is set to 0, suppressing the influence of the explicit pathway on the final classification representation. This supervision is implemented via binary cross-entropy between g_{exp} and the corresponding binary target. This loss is denoted as $\mathcal{L}_{\text{gate}}$.

Hard negative contrastive loss. To enhance the ability of the final representation h_f to distinguish boundary samples, we introduce a supervised contrastive loss. Within each batch, the model uses h_f as the sample representation and constructs positive and negative pairs based on sample labels. For each anchor sample, the cosine similarity between the anchor and all different-class samples in the batch is computed, and the top- k most similar ones are selected as hard negatives for contrastive learning. This targeted selection focuses the contrastive objective on the most confusable sample pairs, rather than wasting gradient signal on easily distinguishable ones. We apply the contrastive loss to h_f rather than h_t because h_f incorporates the injected target-context relationship information; constraining h_f directly ensures that the contrastive objective enhances the discriminability of the final classification representation. This loss is denoted as $\mathcal{L}_{\text{con}}$.

Total loss. The weighted sum of the above losses constitutes the complete training objective of TADI:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_t \mathcal{L}_{\text{tar}} + \lambda_g \mathcal{L}_{\text{gate}} + \lambda_c \mathcal{L}_{\text{con}}, \quad (14)$$

where λ_t , λ_g , and λ_c are the weight coefficients for the respective auxiliary losses. During training, the auxiliary losses employ a progressive warm-up strategy, maintaining lower weights in the early phase and gradually increasing them to the designated values as training progresses, so as to reduce the interference of auxiliary objectives with the primary classification objective during early training.

4. Experiments

4.1. Experimental Setup

Datasets. We conduct experiments on four public hate speech and offensive language datasets covering different languages and task focuses: ETHOS [22] provides English hate/non-hate binary labels for basic effectiveness verification; OLID [30] targets English offensive language with hierarchical annotations including target category; IHC [11] is an English implicit hate corpus that is key to validating whether target-context relationship modeling benefits implicit hate identification; and ToxiCN [20] is a Chinese toxicity dataset used to examine the applicability of TADI beyond English. To ensure comparability, we unify all labels into a coarse binary classification (hate/non-hate),

while retaining the original fine-grained labels for sample grouping in contrastive learning.

Baselines. We select the following baselines for comparison.

- *Large language model baselines:* we employ Qwen3.7-Max [23], GLM-5.2 [29], and DeepSeek-V4-Pro [9] to evaluate general-purpose large language models on the same test sets in a zero-shot setting.
- *Pre-trained encoder baselines:* we fine-tune BERT [10, 7] and RoBERTa [18, 6] for both English and Chinese datasets, to assess the baseline detection capability without additional mechanisms.
- *SharedCon* [1]: a contrastive learning method based on shared semantics between explicit and implicit hate.
- *LAHN* [15]: a method employing label-aware hard negative sampling with momentum contrastive learning.
- *AmpleHate* [17]: a target-aware method that performs detection through explicit target identification and target-context relationship injection.

The above baselines cover general-purpose large language models, pure encoder fine-tuning, sentence-level contrastive learning, hard negative contrastive learning, and target-aware modeling, enabling a comprehensive evaluation of TADI's effectiveness.

Implementation details. We use BERT and RoBERTa backbones for all experiments. Specifically, English datasets are encoded with bert-base-uncased and roberta-base, while Chinese datasets are encoded with bert-base-chinese and chinese-roberta-wwm-ext. All experiments adopt a unified configuration with a maximum sequence length of 256, a batch size of 32, 4 training epochs, a learning rate of 2×10^{-5} , and AdamW [19] as the optimizer. The key hyperparameters of TADI include: contrastive learning temperature $\tau_{\text{con}} = 0.07$, number of hard negatives $k = 16$, target attention temperature $\tau = 1.0$, and injection scaling coefficients for both pathways $\lambda_{\text{imp}} = \lambda_{\text{exp}} = 0.05$. For the large language model baselines, we use a unified task instruction and label space across all datasets with deterministic decoding (temperature = 0) and report results from a single run, while for all encoder-based models we report the mean $\pm$ std over 5 random seeds.

Evaluation metric. We adopt Macro-F1 as the primary metric, as it is more sensitive to class imbalance and provides a more comprehensive reflection of the model's detection capability across different classes.

4.2. Main Results

Table 1 reports the in-dataset test results of TADI and all baseline methods on the four datasets. Fine-tuned methods are run under 5 random seeds with mean and standard deviation reported, while large language model baselines are evaluated on the same test sets.

Table 1

In-dataset main experimental results (Macro-F1, %). Bold values indicate the best performance within each method group.

Method	ETHOS	OLID	IHC	ToxiCN
Qwen3.7-Max	85.40	78.02	74.21	77.14
GLM-5.2	85.22	78.13	75.64	75.37
DeepSeek-V4-Pro	84.23	73.59	70.80	76.12
BERT	75.30±1.66	79.54±0.96	76.99±0.41	80.48±1.15
SharedCon _{BERT}	76.26±1.03	80.16±0.87	76.71±0.65	80.04±0.34
LAHN _{BERT}	70.34±0.72	79.13±0.82	77.68±0.46	80.73±0.54
AmpleHate _{BERT}	76.21±1.43	79.58±1.83	77.68±0.87	79.71±1.05
TADI _{BERT}	84.61±1.02	84.49±0.73	80.73±1.03	82.22±0.55
RoBERTa	75.06±0.72	80.10±0.54	78.04±0.36	81.82±0.19
SharedCon _{RoBERTa}	80.44±2.22	80.25±0.89	78.04±0.28	82.01±0.44
LAHN _{RoBERTa}	75.65±0.36	80.16±1.72	78.34±0.62	82.01±0.59
AmpleHate _{RoBERTa}	77.78±1.27	79.73±0.55	77.13±1.03	81.28±0.59
TADI _{RoBERTa}	87.50±1.21	83.88±1.20	82.40±0.61	83.86±0.36

Across all fine-tuning settings, TADI achieves the best Macro-F1 in all 8 experimental configurations composed of 4 datasets and 2 encoders, demonstrating that its effectiveness does not depend on a specific dataset or encoder and that it can maintain stable detection capability across different languages and label systems. In the comparison with AmpleHate, the target-aware baseline most closely related to the present work, TADI maintains a consistent advantage under both BERT and RoBERTa encoders, and its margin of improvement does not fluctuate drastically across datasets. This stable gap indicates that the implicit target pathway and auxiliary training mechanisms introduced in TADI provide additional discriminative information beyond AmpleHate, whose explicit target modeling still depends on target identification.

The advantage of TADI is particularly notable on IHC, the implicit hate speech dataset. The samples in IHC mostly do not contain overtly offensive words and rely more on the understanding of the target and its contextual relationships. TADI's improvement over all baselines on this dataset is comparable to or even greater than that on the other datasets. This result aligns with the core motivation of this paper: in scenarios where explicit cues are absent, explicit modeling of target-context relationships is key to improving implicit hate speech identification capability.

Furthermore, the performance of the three large language models exhibits pronounced task dependence. They achieve results comparable to TADI on ETHOS, where the label space is simple and expressions are relatively direct, but fall notably behind TADI and most fine-tuned baselines on OLID, IHC, and Chinese ToxiCN. This indicates that, although general-purpose large language models possess strong zero-shot capabilities, task-specific target-aware methods retain irreplaceable advantages on tasks that require fine-grained target-context semantic understanding, such as implicit hate speech identification and Chinese toxicity detection.

4.3. Ablation Study

To verify the contribution of each module in TADI, we design ablation experiments based on TADI_{RoBERTa}. The specific settings include: (1) w/o Target: removing target-related supervision, including attention supervision, category supervision, and gate supervision; (2) w/o Injection: removing the injection of target-context relationship vectors; (3) w/o Explicit: removing the explicit target pathway; (4) w/o Implicit: removing the implicit target pathway; (5) w/o Contrastive: removing the hard negative contrastive loss. Table 2 reports the Macro-F1 results of each setting on the four datasets, with the values in parentheses indicating the change relative to TADI_{RoBERTa} (Full).

The ablation results exhibit a clear hierarchy overall. Across all settings, removing target-related supervision leads to the largest performance degradation, followed by removing relationship injection, and the drop magnitudes of both are substantially higher than those of the remaining settings. This indicates that the primary performance gains of TADI stem from the modeling and utilization of target information, rather than the simple stacking of individual auxiliary modules: without supervisory signals, the model cannot stably learn effective target representations, and merely extracting target representations without injecting their relationship with the context into the classification representation likewise fails to fully exploit the role of target information.

Further examining the explicit and implicit pathways, removing either pathway causes performance degradation, indicating that the two provide complementary target information. Among them, removing the explicit pathway has a larger average impact, reflecting the fact that explicit candidates can provide direct positional priors. Meanwhile, the impact of removing the implicit pathway is notably higher on IHC than on the other datasets. Considering that IHC focuses on implicit hate speech expressions, this phenomenon demonstrates that the implicit target extraction mechanism plays a more prominent role when explicit cues

Table 2

Ablation study results based on $\text{TADI}_{\text{RoBERTa}}$ (Macro-F1, %). Bold values indicate the best performance. Values in parentheses indicate performance drops compared with the full model.

Setting	ETHOS	OLID	IHC	ToxicCN
$\text{TADI}_{\text{RoBERTa}}$ (Full)	87.50	83.88	82.40	83.86
w/o Target	72.51 (−14.99)	79.79 (−4.10)	77.89 (−4.51)	80.77 (−3.09)
w/o Injection	75.35 (−12.14)	80.51 (−3.38)	78.78 (−3.62)	81.49 (−2.37)
w/o Explicit	84.26 (−3.23)	81.57 (−2.31)	80.02 (−2.38)	82.04 (−1.82)
w/o Implicit	86.15 (−1.35)	83.21 (−0.67)	80.74 (−1.66)	83.09 (−0.77)
w/o Contrastive	85.82 (−1.67)	82.91 (−0.98)	81.16 (−1.23)	83.13 (−0.73)

are absent, which is also consistent with the motivation of this paper regarding implicit target modeling. In comparison, the degradation caused by removing contrastive learning is the smallest across all datasets but consistent in direction, indicating that the hard negative contrastive loss is not the primary source of TADI's performance gains but rather further enhances the discriminability of boundary samples on the basis of target-aware representations.

4.4. Cross-Dataset Generalization

To evaluate the generalization capability of TADI under distributional shift, we conduct bidirectional cross-dataset experiments between ETHOS and IHC: the model is trained exclusively on the source dataset and tested directly on the target dataset. This combination is selected primarily based on semantic consistency of the task, as both datasets center on hate speech detection and are complementary in expression form (ETHOS tends toward explicit expressions, while IHC tends toward implicit expressions). In contrast, the positive class in OLID encompasses a large volume of offensive or insulting expressions whose label semantics differ from those of hate speech; including it in the experiment would cause performance variation to be simultaneously affected by domain differences and label definition discrepancies, making it difficult to effectively assess generalization capability. Considering the different scales of the two datasets, balanced sampling is applied during training to mitigate the impact of sample size inconsistency. We do not include ToxicCN in this setting because cross-lingual transfer introduces language shift in addition to dataset shift, which is beyond the scope of this evaluation. The results are presented in Table 3.

TADI achieves the best results in all settings composed of two transfer directions and two encoders, demonstrating its applicability even when the training and test sets originate from different sources. Notably, the improvement of TADI over AmpleHate, the most closely related target-aware baseline, is generally larger in the cross-dataset setting than in the in-dataset experiments, with the ETHOS → IHC direction being particularly pronounced. This suggests that when the distributions of the training and test sets are inconsistent, approaches that rely solely on explicit target candidates are more susceptible to the influence of dataset differences, whereas TADI, by modeling potential target information

Table 3

Cross-dataset generalization results (Macro-F1, %). Bold values indicate the best performance within each encoder group.

Method	ETHOS → IHC	IHC → ETHOS
BERT	60.25	67.07
SharedCon _{BERT}	59.06	68.01
LAHN _{BERT}	55.11	58.41
AmpleHate _{BERT}	59.39	68.68
$\text{TADI}_{\text{BERT}}$	66.48	71.78
RoBERTa	62.45	66.61
SharedCon _{RoBERTa}	63.04	66.58
LAHN _{RoBERTa}	56.76	59.97
AmpleHate _{RoBERTa}	63.26	67.17
$\text{TADI}_{\text{RoBERTa}}$	71.25	73.66

from within the text through implicit target extraction, is able to learn more transferable discriminative cues.

Nevertheless, the cross-dataset results of all methods remain substantially lower than the in-dataset test results overall. For example, $\text{TADI}_{\text{BERT}}$ achieves an in-dataset Macro-F1 of 80.73% on IHC, while the ETHOS → IHC transfer result drops to 66.48%. This indicates that differences in linguistic expression, label distribution, and sample composition across datasets remain important factors constraining detection performance. Overall, target-context relationships can provide useful transferable cues for cross-dataset detection, but cross-dataset hate speech detection remains a challenging task.

4.5. Case Analysis

To further examine whether TADI captures target-related discriminative cues, we visualize the word-level target attention derived from the target extraction module. The token-level attention weights are computed via Eq. (4) and aggregated to the word level for readability. It should be noted that the attention visualized here originates from TADI's internal target extraction module, not from the Transformer encoder's own self-attention. We select three representative cases from the correctly classified samples of $\text{TADI}_{\text{RoBERTa}}$ on the IHC and ETHOS test sets, corresponding to implicit hate, explicit hate, and non-hate scenarios, respectively. The results are shown in Fig. 3. The figure also reports two metrics computed from word-level attention: W-Ent. is the

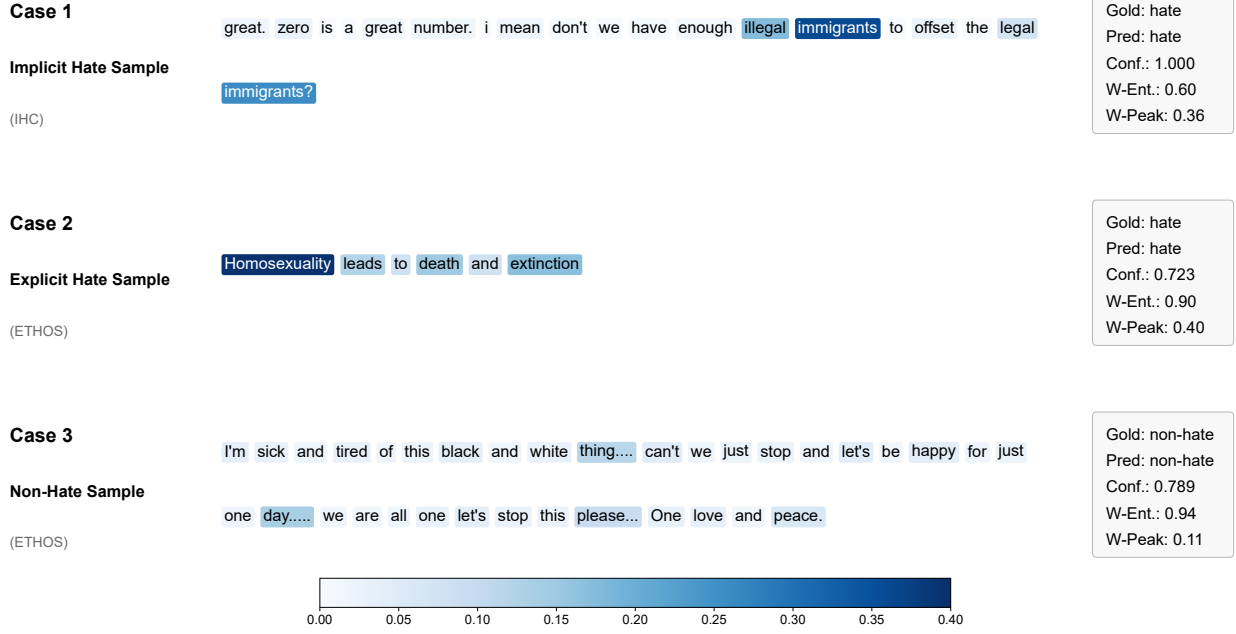


Figure 3: Visualization of word-level target attention from TADI_{RoBERTa} on three representative cases.

normalized attention entropy, with values in $[0, 1]$, where values closer to 0 indicate more concentrated attention and values closer to 1 indicate more dispersed attention; W-Peak is the maximum word-level attention weight, reflecting the strongest degree of focus on a single word.

In the implicit hate sample (Fig. 3, Case 1, from IHC), the text contains no overtly abusive words but instead expresses exclusion of immigrant groups through a rhetorical question. The model assigns higher weights to words such as *immigrants*, *illegal*, and *legal* (W-Ent. = 0.60), with a relatively concentrated attention distribution. This demonstrates that TADI can still capture potential targets of attack and their contextual cues from within the text even in the absence of explicit abusive words. In the explicit hate sample (Case 2, from ETHOS), the model attends to the target word *Homosexuality* while also distributing weight to negative contextual words such as *extinction* and *death*, with an attention entropy higher than that of Case 1 (W-Ent. = 0.90). This contrast indicates that for explicit hate samples, the model not only identifies the target itself but also attends to multiple contextual words that constitute the offensive semantics, reflecting the joint modeling of target-context relationships.

In contrast to the above two types of samples, in the non-hate sample (Case 3, from ETHOS), although group-related words such as *black* and *white* appear in the text, the attention distribution is notably more dispersed (W-Ent. = 0.94, W-Peak = 0.11), without strongly focusing on any specific word. This demonstrates that TADI does not simply rely on group-related words for judgment, but instead makes

its non-hate determination by combining the semantics of peace and unity expressed in the context. Taken together, the three cases show that TADI can capture potential targets of attack in implicit hate samples, simultaneously attend to both the target and offensive context in explicit hate samples, and avoid treating group-related words as direct evidence of hate in non-hate samples. This further indicates that target attention can provide interpretable discriminative cues for TADI's target-context relationship modeling.

5. Conclusion

In this paper, we proposed TADI, a target-aware framework for hate speech detection that models target information through implicit and explicit pathways and injects target-context relationships into the classification representation. Instead of relying only on sentence-level representations, TADI uses attention-based implicit target extraction and explicit target candidate aggregation to capture complementary target cues. The gated residual injection mechanism further allows the model to incorporate target-context relation information while preserving the original task representation.

Experiments on four hate speech and offensive language datasets with two pre-trained encoders demonstrate the effectiveness of TADI in both in-dataset and cross-dataset settings. The main results show that TADI consistently improves over encoder-based baselines and existing contrastive or target-aware methods. The ablation study further confirms that target modeling and target-context relation injection are the primary sources of performance gains, while the hard

negative contrastive objective provides additional improvements for boundary samples. The cross-dataset results indicate that target-context relationship modeling can provide useful transferable cues under distributional shift.

Despite these results, several limitations remain. First, although TADI improves cross-dataset performance over the baselines, its transfer results still remain lower than in-dataset results, indicating that differences in annotation schema, label distribution, and expression style continue to constrain generalization. Second, the explicit pathway relies on target candidate information derived from surface-form matching, which may introduce noise when candidate terms are ambiguous or when targets are expressed indirectly. Third, the current framework focuses on textual hate speech and does not address multimodal scenarios where images, memes, or videos may carry important hateful signals.

Future work will explore more robust target discovery methods with weaker dependence on surface-form matching. Incorporating external knowledge or retrieval-augmented mechanisms may help resolve culturally coded, metaphorical, or context-dependent targets. Extending TADI to multimodal hate speech detection and broader cross-lingual settings is also an important direction for improving the reliability and applicability of target-aware hate speech detection.

CRedit authorship contribution statement

Shan Jin: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – original draft. **Xiaochao Fan:** Conceptualization, Resources, Funding acquisition, Supervision, Project administration, Writing – review & editing. **Daoxiang Cheng:** Investigation, Validation, Data curation, Writing – review & editing. **Palidan Tuerxun:** Resources, Supervision, Project administration, Writing – review & editing. **Shaotao Hu:** Investigation, Validation, Visualization, Writing – review & editing. **Zhenzhen He:** Investigation, Validation, Data curation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No new dataset was created in this study. The datasets analyzed in this study are publicly available from the original data providers cited in the paper. The source code is publicly available at <https://github.com/jin-shan/tadi>.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT to improve the grammar and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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